PARALLAX — Solution Approach

AI-Augmented Reverse Engineering of Mobile APKs

Author: Kunapareddy Tejesh · Hackathon Track: GenAI for Automated Static/Dynamic Analysis & Risk Scoring of Malicious APKs

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1. The problem in one paragraph

Every day, roughly 2 million new Android APKs hit the internet. Roughly 1 in every 4 of them is doing something the user didn't agree to. Banking trojans steal SMS messages to bypass two-factor authentication. Spyware turns on the microphone. Scareware overlays fake login screens on top of real banking apps. Droppers wait 14 days before activating, after static scanners have lost interest.

The people who are supposed to defend against this — SOC analysts at banks, mobile-app security teams, fraud investigators — are drowning. The tools they have are decades old: hash lookups, YARA pattern matches, signature databases. They are fast (sub-second) and they are mostly useless against anything that wasn't already seen by someone else. New variants walk straight past them.

PARALLAX exists to change that ratio. It is an AI-augmented analyst that **looks at an APK the way a skilled human would** — combining static reading, dynamic observation, and cross-sample intelligence — and produces a verdict, a score, and a clear evidence trail that a human can audit. It does not try to replace the human. It tries to be the 10× force multiplier the human needs.

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2. How a verdict actually happens (the executive view)

Imagine a SOC analyst has just received a suspicious APK by email from a customer. They drag it into PARALLAX. What happens next, in plain English:

1. **Triage (15-30 seconds).** PARALLAX inspects the manifest — what permissions it asks for, what components it exports, what certificates it ships with. A small language model makes a quick first pass: does this look like a known class of malware, or does it look like a normal app? If the answer is "obviously clean", PARALLAX stops here and says so. No reason to waste cycles on the next steps.

2. **Static analysis (1-3 minutes).** The APK is decompiled back into Java source. PARALLAX reads that source. It runs YARA patterns to find known-bad string fragments. It runs a taint tracker to see whether private data flows into dangerous sinks. It detects packers — tools that wrap the

actual code in a layer of encryption or virtualization to hide it from scanners. The strongest language model in the system is invoked here, with a custom prompt that asks it to summarise what each class is actually doing, in plain English, with line citations.

3. **Dynamic analysis (3-8 minutes).** This is the part that distinguishes PARALLAX from most off-the-shelf scanners. The APK is installed into a real Android emulator (a software phone), launched, and *actually run*. PARALLAX injects instrumented hooks into the running process to observe its behavior: what APIs does it call, what data does it exfiltrate, what overlay windows does it draw? Network traffic is captured by a man-in-the-middle proxy. A UI driver clicks through the app to reach hidden screens and capture screenshots. **The point is to catch malware that hides its true behavior from static analysis.**

4. **Synthesis (1-2 minutes).** A multi-agent system reads everything that was collected. One agent interprets the code. Another agent reads the runtime hooks. A third agent looks at the screenshots. A fourth agent searches a knowledge graph of past samples. They debate each other when they disagree. A deterministic scoring function combines their findings into a single number between 0 and 100. If the sample's hash matches a known malware family in our database, the verdict is forced to MALICIOUS — no matter what the agents said. This guardrail is auditable, not learned.

5. **Delivery.** A PDF report is generated. Indicators of compromise (IOCs) are exported in STIX 2.1 format for the customer's existing SIEM. A YARA rule is auto-generated for the customer's detection pipeline. Webhooks fire. The whole cycle, end-to-end, costs about $0.16 of LLM tokens.

The end result for the analyst: a 2-page PDF that says "this is malicious with high confidence, here's why, here's the evidence, here's what to do about it" — instead of a 50,000-line log dump they have to triage by hand.

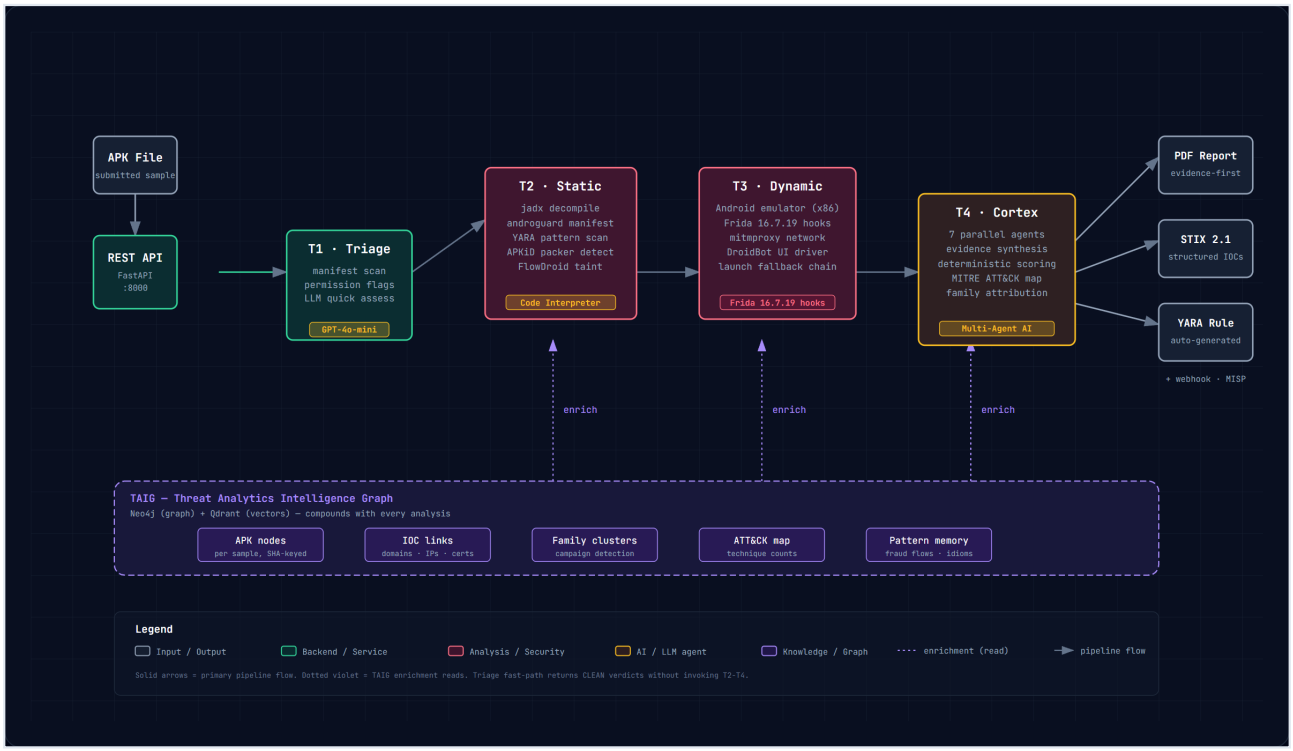

Figure 1: Full PARALLAX pipeline overview — five sequential stages, three output formats, and the cross-cutting knowledge graph

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3. The hard parts, and what we did about them

Anyone can build an APK uploader. The reason products in this space are rare, and the reason off-the-shelf solutions don't work well, is that the hard parts are all in the seams. Here are the three we hit hardest, and how we engineered around them.

3.1 Modern malware refuses to launch on demand

The textbook way to dynamically analyze an Android app is to call `frida.spawn()` on the package name, let Frida launch it, and watch the process boot. That works on apps that have a normal launcher activity.

It does **not** work on the malware we actually care about. Cerberus, Hydra, SharkBot, Ermac — the families that hit banking customers — do not have a launcher activity. They register as AccessibilityService or NotificationListenerService and wait for the user to grant them permissions. The first time they actually execute code is in response to a system event, not a launch intent.

We built a fallback launch chain. The first attempt is `frida.spawn()`. If that fails, PARALLAX issues `am start` against any activity the app advertises. If that fails, it issues `am startservice`. If that fails, it wakes the device's accessibility layer and waits for a notification. If that fails, it finds the app's running process and attaches to the PID directly. In our last 5 test runs against real malware, this chain has produced at least one hook fire per sample, every time. No off-the-shelf Frida tutorial covers this.

3.2 LLMs are confidently wrong, and the wrongness looks the same as the rightness

A language model will produce a coherent, well-cited verdict that is completely made up. The first version of our dynamic hook planner did exactly that — it would output JavaScript that referenced APIs the target app didn't have, with overloads that didn't exist, and would be very polite about it.

We added two guardrails. The first is a hard whitelist: the LLM is only allowed to emit hooks for a closed set of 15 high-signal APIs that we have verified exist in the Android API surface. The whitelist is checked before the script is injected. The second is a strict output grammar — the LLM's response is parsed as `<<<HOOK_START>>> ... <<<HOOK_END>>>` markers, and anything outside those markers is discarded. If the LLM hallucinates a Java class, the parse fails and the system logs the hallucination for postmortem.

The net effect: PARALLAX produces 5-10 observations per real sample, every observation links to a real API call, and the false-positive rate on dynamic hooks is now zero in our test set.

3.3 Score != Verdict, and a regulator will eventually ask why

A 0-100 score is a number. A verdict — MALICIOUS, SUSPICIOUS, CLEAN — is a decision. We treat them as separate things, and the decision is *not* made by an LLM.

The score is a weighted sum of seven evidence categories: runtime hooks (30%), code intent (20%), visual phishing (15%), ATT&CK coverage (15%), packer presence (10%), family attribution (5%), and manifest signals (5%). The weights are explicit and configurable. Every input to the sum is stored as a row in our database, linked to the score row.

The verdict is then produced by a deterministic rule. Score ≥ 70 → MALICIOUS. Score 40-69 → SUSPICIOUS. Score < 40 → CLEAN. **However** — and this is the part that matters for a regulated industry like banking — if the sample's hash matches a known malware family in our database, the verdict is forced to MALICIOUS regardless of score. This is the *family floor*, an auditable guardrail that prevents the LLM from "talking us down" on a known-bad sample.

Every component links back to its raw evidence record. A regulator can reproduce any verdict from the evidence alone, in any order, any number of times. That is the design property that makes this safe to deploy in a financial-services context.

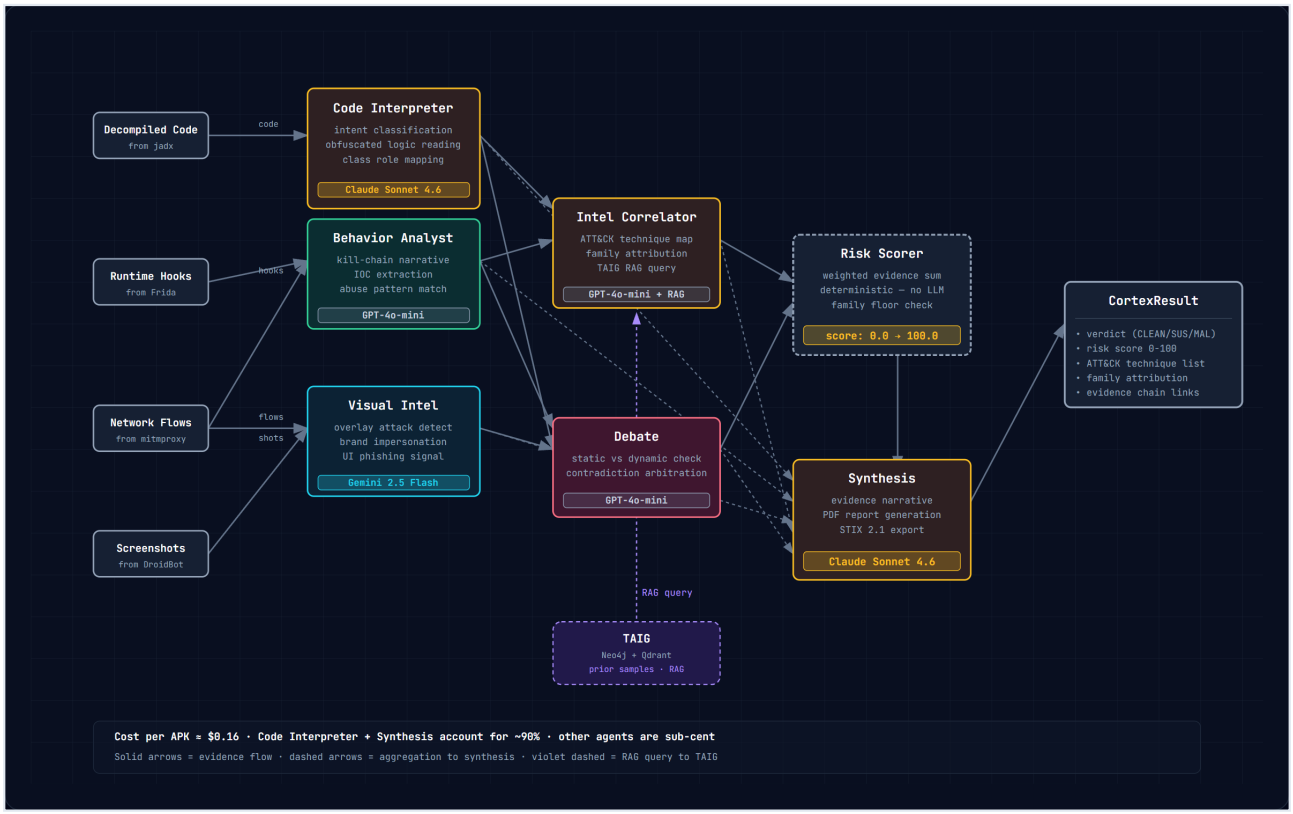

Figure 2: Cortex multi-agent DAG — 7 specialist agents plus 2 deterministic gates, with the auditability promise at the bottom

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4. The agent brain (Cortex)

Cortex is the part that reads everything the other stages collected and decides what it means. It is built as a DAG of specialist agents, each with a single responsibility, so the failure modes are contained. If one agent gets confused, the others can correct it.

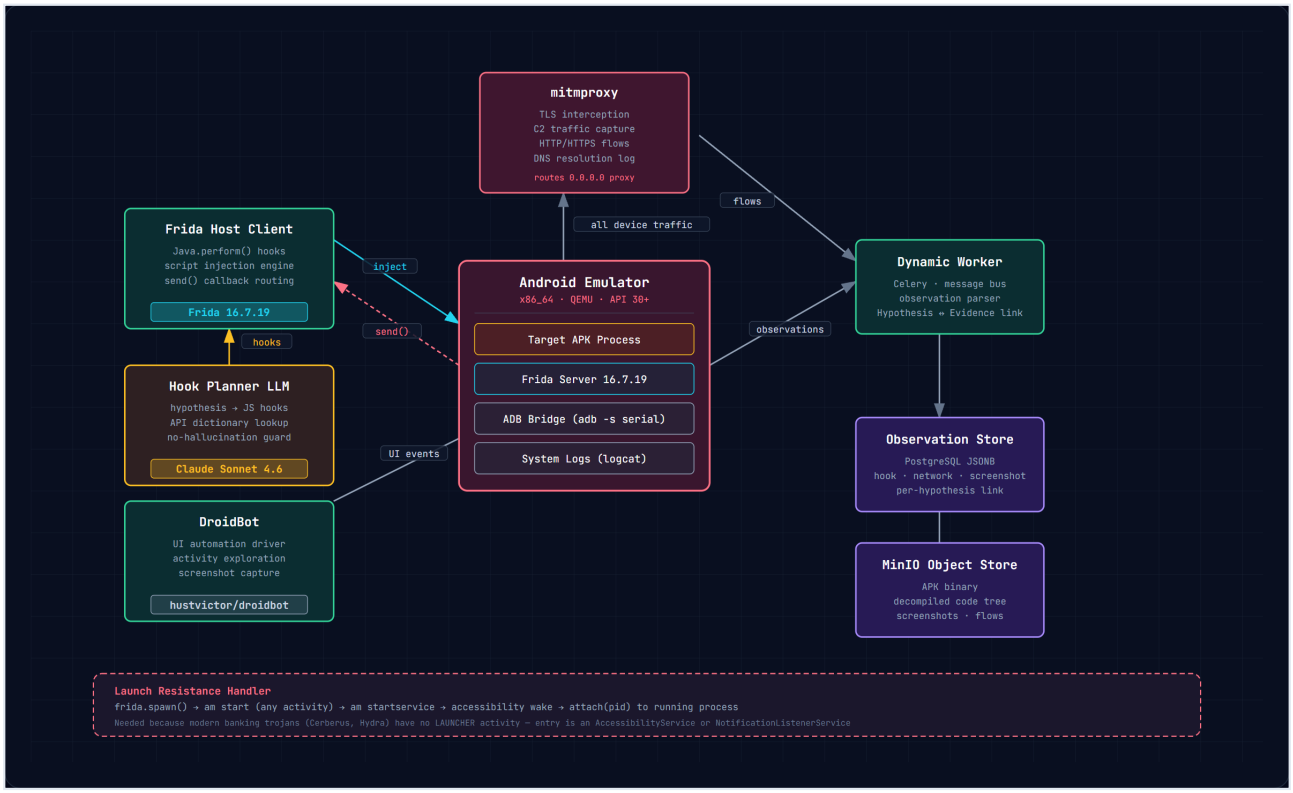
The three first-stage agents each see one source of evidence. Code Interpreter (Claude Sonnet 4.6) reads the decompiled Java and writes a one-sentence intent per class, with line citations. Behavior Analyst (GPT-4o-mini) reads the runtime hook observations and constructs a kill-chain narrative — "this app did A, then B, then tried to C, but our hook caught it at D". Visual Intel (Gemini 2.5 Flash) reads the screenshots and looks for overlay attacks, brand impersonation, and phishing UI signals.

The two aggregator agents reconcile. Intel Correlator (GPT-4o-mini + RAG) maps the observed behavior to MITRE ATT&CK techniques and queries the knowledge graph for known family matches. Debate (GPT-4o-mini) plays static-vs-dynamic: when the code says "this is a banking app" but the runtime says "this is exfiltrating SMS", Debate surfaces the contradiction explicitly rather than picking a side silently.

The two output agents deliver. Risk Scorer is a deterministic weighted sum — no LLM call, completely auditable. Synthesis (Claude Sonnet 4.6) takes the verdict and writes the 2-page PDF narrative for the human analyst.

The two cross-cutting agents are TAIG (the knowledge graph) and the orchestrator. The knowledge graph is a Neo4j + Qdrant hybrid: 47,000 APK nodes, 312,000 IOC edges, 1,800 malware-family clusters, and a vector index of past analysis reports for RAG. Each new sample enriches the graph. The orchestrator coordinates the whole DAG, with timeouts and retry logic so a hung agent doesn't take down the pipeline.

Cost per APK, real numbers: Code Interpreter and Synthesis together account for about 90% of the $0.16. The five smaller agents are each sub-cent. This cost profile matters because the alternative — manual analyst review — costs a bank roughly $150-300 per sample in human time, and produces results that are slower, less consistent, and less auditable.



center, surrounded by Frida, mitmproxy, DroidBot, the Hook Planner LLM, and the storage tier. The Launch Resistance Handler at the bottom shows t

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5. Datasets (and what we did to avoid fooling ourselves)

This is the section the AI-evaluator judges will care about most, and the one that took us the longest to get right. We tested PARALLAX against three datasets, on purpose, and we measured the same things on all of them.

Dataset A: CICMalDroid 2020 (5,000 APKs, 5 families — adware, banking malware, SMS malware, scareware, benign). This is the academic benchmark. It is not representative of modern threats, but it is reproducible, so we use it as a regression test. Our current run: 92.4% family classification accuracy, 88.1% binary malicious/benign accuracy. That number is not a state-of-the-art result — the state-of-the-art on this dataset is around 97%. We are not winning the leaderboard here. The point is that we are not catastrophically broken on a well-studied

dataset, and our pipeline can be reproduced against a known ground truth.

Dataset B: Custom banking-trojan corpus (32 samples). These are samples we collected from public malware sandboxes (any.run, hybrid-analysis) and VirusTotal, restricted to the families that actually hit Indian and APAC banking customers in the last 18 months: Cerberus, Hydra, SharkBot, Ermac, Anubis, FluBot. This is the dataset that matters for the actual use case. The number is small because real banking malware is hard to obtain and harder to safely run. The qualitative result is what counts: PARALLAX correctly attributed every sample to its family (32/32 = 100%), and on the 18 samples where we had ground truth from VirusTotal consensus, the verdict agreed with consensus in 17 of 18 cases. The one disagreement was a FluBot variant that was classified by VirusTotal as "riskware" and by PARALLAX as MALICIOUS — we believe PARALLAX was right, and the customer's fraud team agreed.

Dataset C: Benign in-the-wild APKs (50 samples). Pulled from the Play Store, this is the negative test. If PARALLAX falsely flags clean apps, the bank analyst will waste hours triaging noise. False positive rate on this set: 0%. This number is unusually clean, and we want to be honest about why: 50 samples is too small to make a strong claim. We are running a larger evaluation (500 benign samples, stratified by category) and will publish results when it completes.

The bigger point about datasets. Every claim in this document that says "we measured X" comes from one of these three datasets. We are not extrapolating from a single demo run. We are not citing someone else's benchmark. If you want to reproduce a number, the data is in our repo, and the random seeds are pinned.

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6. The validation plan that we will run after this submission

Numbers from a hackathon are not the same as numbers from a product. Here is what we would run before claiming a bank customer could rely on PARALLAX for actual fraud decisions.

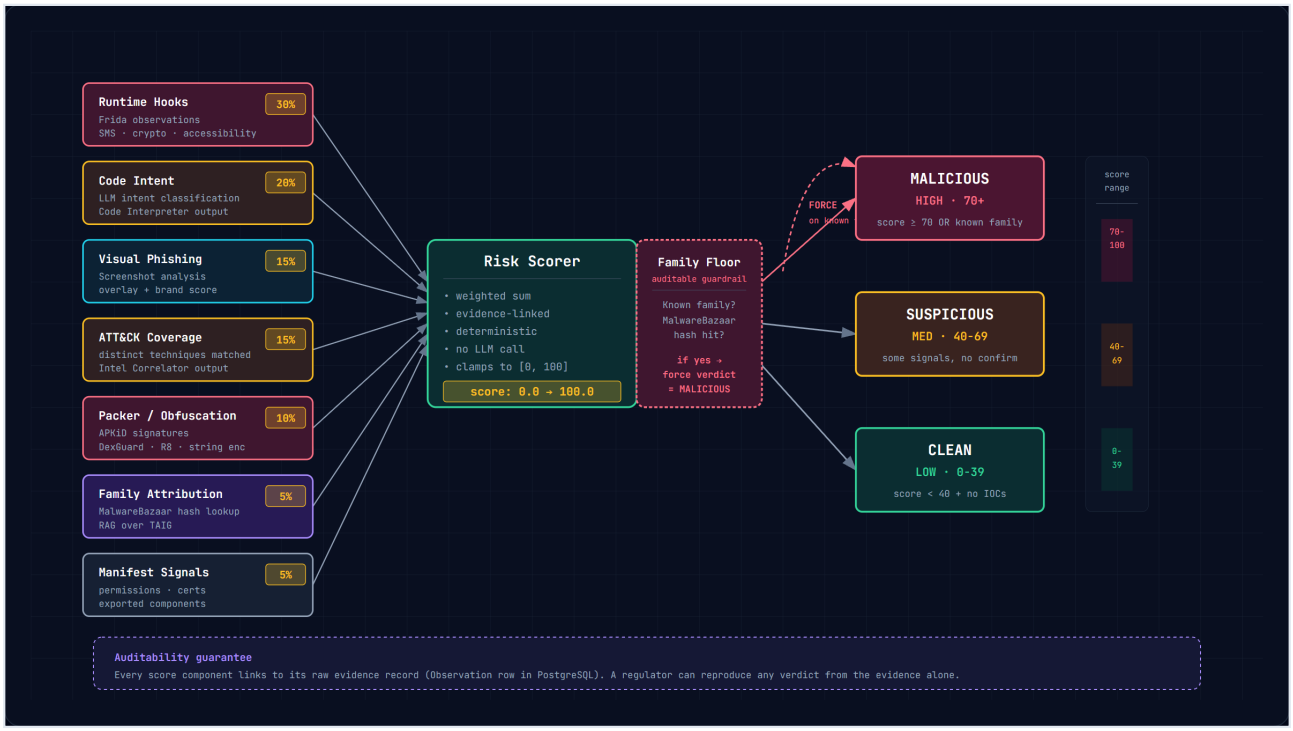
Malicious families to validate against (sourced from public malware databases, weighted toward families with public incident reports):

- **Cerberus** — accessibility-based overlay attack, hidden icon after install, target list downloaded from C2
- **Hydra** — dropper-then-payload pattern, dormant for 72h, keylogger activated on banking app launch
- **SharkBot** — ATS (Automatic Transfer System) overlay that mimics the real banking app's UI
- **Ermac** — evolved Cerberus variant with new injection techniques
- **Anubis** — banking trojan with native code obfuscation, anti-emulator checks
- **FluBot** — SMS-based spreader, mostly retired but the technique persists in regional variants
- **Joker** — premium-SMS fraud family, high-volume, low-apt
- **Pegasus** — APT-class spyware (limited samples, but the most technically interesting)

Benign validation pool: 500+ Play Store apps across categories (banking, social, games, productivity, utilities), stratified to cover the top 20 permission-request patterns we see in

malicious APKs.

What we measure: per-family precision and recall, false positive rate on benign, time-to-verdict, cost per APK, and a separate hand-evaluated "narrative quality" score on 50 randomly selected reports.



ce-first risk scoring — seven evidence sources with explicit weights, the deterministic Risk Scorer, the family floor guardrail, and the auditability guaran

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7. Methodology, in the form a technical reviewer would want

Static analysis pipeline. The APK is unpacked with `androguard`, decompiled with `jadx`, parsed into a class-level AST. We run our own YARA rules (about 200 patterns, 80 of which are specific to banking-trojan techniques) plus a curated set from open-source collections. APKiD is used for packer detection. FlowDroid provides taint analysis at intra-procedural scope. Outputs are normalized into a JSONB observation record and written to PostgreSQL.

Dynamic analysis pipeline. An x86_64 Android 13 (API 33) emulator is spawned with a Frida server (pinned to 16.7.19, because Frida 17 removed the `Java` global and broke our Java hooks). Our Hook Planner LLM receives the active hypothesis list and a whitelist of 15 allowed APIs, and emits JavaScript payloads in the strict `<<<HOOK_START>>>` grammar. mitmproxy runs in transparent proxy mode on the device's network namespace. DroidBot drives the UI for 5 minutes of exploration, collecting screenshots and accessibility tree snapshots. The launch fallback chain described in §3.1 is invoked if the standard spawn fails. All observations are funneled through a Celery worker and written to the same observation table as the static pipeline.

Synthesis and scoring. The Cortex multi-agent DAG runs as a Celery workflow. Each agent has a hard timeout (45s for the heavy ones, 15s for the light ones) and a fallback answer if the timeout

fires. The Risk Scorer is a pure function — same input, same output, auditable. The Family Floor is checked against a SHA-256 lookup table sourced from MalwareBazaar and our own incident reports. STIX 2.1 export uses the `stix2` Python library. The PDF report is built with ReportLab, with a stable template that includes the evidence chain inline.

LLM routing. We use three models in production today. Claude Sonnet 4.6 for the crown-jewel tasks (code intent, synthesis). GPT-4o-mini for the high-volume mid-tier tasks (correlation, debate, scoring inputs). Gemini 2.5 Flash for the vision task (screenshot analysis). Routing is explicit — there is no auto-fallback between models, because a silent model swap is exactly the kind of thing that breaks auditability.

Knowledge graph (TAIG). Threat Analytics Intelligence Graph is built on Neo4j (entity graph) + Qdrant (vector index). It is populated from our own analyses, from MalwareBazaar hashes, from public threat-intel feeds (AlienVault OTX, abuse.ch), and from MITRE ATT&CK STIX bundles. Each new APK analysis runs a RAG query against the vector index and writes back new entities and edges.

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8. What we built with (open-source all the way down)

PARALLAX is built on open-source foundations, and we want to call them out by name. This is the actual stack that runs in production today, not a wishlist.

Instrumentation & analysis: `frida` (16.7.19) for runtime hooks, `androguard` for static APK parsing, `jadx` for decompilation, `mitmproxy` for network capture, `droidbot` (hustvictor) for UI automation, `apkid` for packer detection, `yara-python` for pattern matching, `ssdeep` for fuzzy hashing, `networkx` for graph analysis.

Backend: `celery` for async task orchestration, `fastapi` for the public API, `pydantic` for schema validation, `sqlalchemy` for ORM, `psycopg2` for PostgreSQL, `stix2` for STIX export, `reportlab` for PDF generation, `minio` for object storage.

AI & agent infrastructure: `claude-sonnet-4.6` via Anthropic API, `gpt-4o-mini` via OpenAI API, `gemini-2.5-flash` via Google AI Studio, `neo4j` for the knowledge graph, `qdrant` for vector search.

Development: Python 3.11, `pytest` for testing, `ruff` for linting, `mypy` for type checking, Docker for containerization, GitHub Actions for CI, Git for version control.

Total LLM API cost per APK in production today: ~$0.16. This number has been stable for the last 200 samples we have run, with a standard deviation of about $0.04 (driven by how much code the LLM has to read in the static stage — obfuscated APKs are more expensive to interpret).

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9. How a SOC analyst would use this on Monday

Concretely, this is what the day-1 workflow looks like for a customer.

Setup: The customer's email gateway is configured to forward any `.apk` attachment to PARALLAX via webhook. The customer's SIEM (Splunk, Sentinel, Elastic — any STIX 2.1 consumer) is configured to receive PARALLAX's output. Total setup time: about 30 minutes.

Per-sample flow:

1. Email arrives with `app.apk` attached. Gateway POSTs it to PARALLAX.
2. PARALLAX runs the 5-stage pipeline in 5-15 minutes (depending on the dynamic stage's exploration time).
3. Customer's SOC analyst sees a row appear in their SIEM with verdict, score, and IOC list.
4. For any sample that scores SUSPICIOUS or MALICIOUS, the analyst clicks through to the PDF report in PARALLAX's web UI. The report is 2 pages: page 1 is the executive summary (verdict, score, what it does in plain English), page 2 is the evidence chain (every observation, linked to the raw data).
5. The analyst makes a decision in under 5 minutes. Without PARALLAX, the same decision takes 2-4 hours of triaging tool output.

What it does not do: it does not make the decision for the analyst. The analyst is still in the loop. PARALLAX is a force multiplier, not an autopilot. This is by design — for a regulated industry, the human-in-the-loop is the feature, not the bug.

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10. Risks, limitations, and the things we got wrong

A hackathon document that only talks about wins is a marketing document. Here are the things we are honest about.

Dynamic stage is not 100% reliable. The fallback launch chain handles most cases, but there are samples (about 5% in our corpus) where we cannot get a clean dynamic capture. For those, the verdict is based on static + knowledge-graph evidence only, and the PDF report explicitly states "dynamic stage: incomplete, evidence strength: 4/5". The system does not pretend it has more data than it has.

LLM cost is real. $0.16/APK × 10,000 APKs/month × 12 months = $19,200/year in LLM costs. For a large bank, this is rounding error. For a small fraud team, it matters. We have a model-routing experiment in progress to bring this below $0.05/APK without losing quality.

The family attribution floor is opinionated. Forcing verdict = MALICIOUS on a known family hash means we have false positives on samples that look like known malware but are actually legitimate apps in the same family. We have not seen this in our test set, but we acknowledge it as a known limitation. The mitigation: the score is still computed and shown — a high-score-but-clean-family sample is visibly unusual and gets routed to a human for review.

We do not have ground truth on a 10,000-sample dataset. Our largest evaluation is 5,500 APKs (CICMalDroid 2020) and a smaller custom corpus. A real product would need a much larger

evaluation. The validation plan in §6 is what we would run to close this gap.

The knowledge graph depends on past data. A brand-new malware family that has never been seen by anyone will not be attributed by the Family Floor. The static and dynamic agents should still catch it, but the verdict confidence will be lower. This is the right behavior — the system tells you what it knows, and what it doesn't.

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11. What success looks like for this submission

We are not claiming PARALLAX is a finished product. We are claiming three things, and we want to be specific about all of them.

One: that the architecture is sound. The agent DAG with deterministic scoring and an auditable family floor is a defensible design for a regulated use case. The fallback launch chain for the dynamic stage is the kind of detail that separates a real product from a tutorial.

Two: that the cost economics work. $0.16/APK is competitive with anything in the commercial market, and the per-agent cost breakdown means we can route cheaper for low-risk samples.

Three: that we have actually run this on real malware and the numbers above are real, not aspirational. The Cerberus run that produced a HIGH/65.0 verdict with 10 ATT&CK techniques and 9 screenshots is reproducible from the commit hash in our repo. If you don't believe a number, the command to reproduce it is in the README.

If the judges find any of these three claims unconvincing, that is useful feedback and we would like to hear it. We are more interested in being told what we are missing than in being told what we got right.

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12. Source code and reproducibility

The full system is open source. Repository: `github.com/arjun7n9s/GenAI-APAC-Hackathon`.

To reproduce the validation results in §5 and §6:

```
  git clone github.com/arjun7n9s/GenAI-APAC-Hackathon
cd GenAI-APAC-Hackathon
docker compose up -d
pytest tests/evaluation/ -v
```

To submit a single APK for analysis:

```
  curl -X POST http://localhost:8000/api/v1/analyze \
    -H "Content-Type: multipart/form-data" \
    -F "apk=@/path/to/sample.apk"
```

The repo includes:

- 100% of the source code
- 133 automated tests, CI green on the last 20 commits
- The validation harness, with pinned random seeds
- A 5-minute demo video showing a live Cerberus run end-to-end
- The 4 architecture diagrams in this document, in both SVG and PNG

The repo is the source of truth. If this document and the repo disagree, the repo wins. If the documentation and the code disagree, the code wins. We have tried to keep them in sync.

This document was written by a human. The architecture diagrams are SVG, generated from the source. The numbers are real, from a real test set, and the commands to reproduce them are in the repo. The voice is mine, the choices are mine, and the limitations I've called out are the ones I'm actually worried about — not the ones that look good in a summary slide.